

Assessing predictions

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- HW 2 due tonight

- HW 3 { HW 4 posted

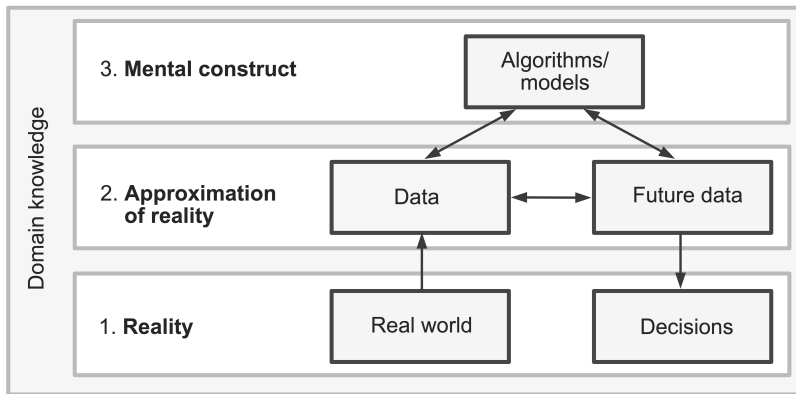
HW 3: Due next week

HW 4: Due in 2 weeks ← Data analysis,
Short report

- Don't forget AI statement!

Model assessment

Essentially, all models are wrong, but some are useful. (Box and Draper, 1987)



Today: Is our good at *predicting* the response?

Warmup

Work on the warmup activity (handout).

Types of prediction

Regression: estimate $\mu_i = E[y_i | x_i]$

E.g. $y_i \sim \text{Bernoulli}(p_i) \rightarrow \hat{p}_i \in (0,1)$

Classification: estimate y_i

($\hat{y}_i$ has the same support as y_i ,
 $\hat{\mu}_i$ does not have to)

E.g. $y_i \sim \text{Bernoulli}(p_i) \rightarrow \hat{y}_i \in \{0,1\}$

General principle: predictions from regression can be converted into classification, but not the other way around!

Assessing predicted probabilities

$$Y_i | \mathbf{x}_i \sim \text{Bernoulli}(p_i)$$

$$\log \left(\frac{\hat{p}_i}{1 - \hat{p}_i} \right) = \mathbf{x}_i^T \hat{\beta}$$

Question: How do we tell if $\hat{p}_i$ is a "good" estimator of p_i ?

$$\text{Likelihood} : L = \prod_{i=1}^n f(y_i | \hat{p}_i) = \prod_{i=1}^n \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i}$$

want big $\nearrow$

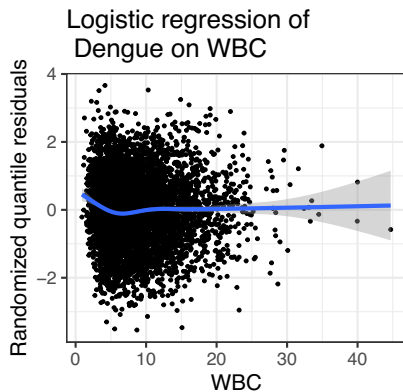
$$\text{Deviance} = -2 \log L = -2 \sum_{i=1}^n [y_i \log(\hat{p}_i) + (1-y_i) \log(1-\hat{p}_i)]$$

$\uparrow$
want small

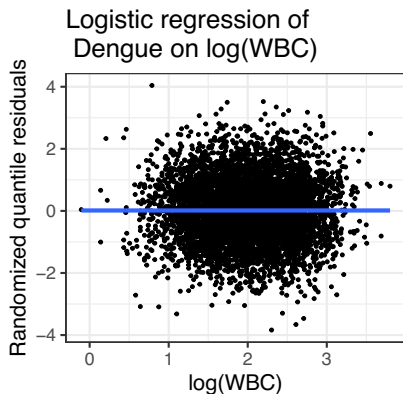
$$\text{Brier score} : \sum_{i=1}^n (y_i - \hat{p}_i)^2$$

$\uparrow$
want small

Assessing predicted probabilities



deviance 5529.825



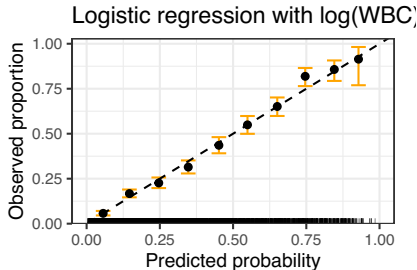
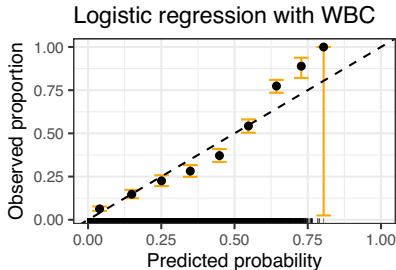
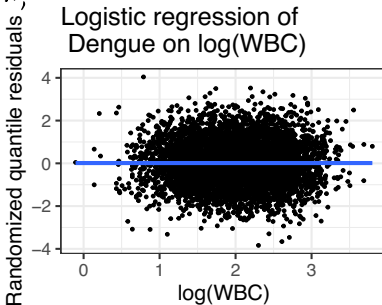
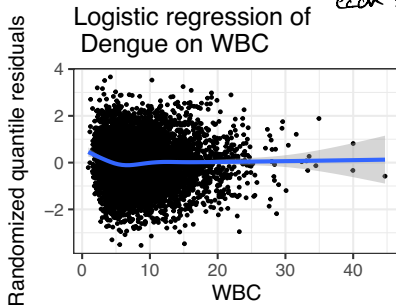
deviance 5410.832

↑
smaller!

Question: Does using WBC vs. $\log(\text{WBC})$ make a difference for the quality of our predictions?

Calibration plot

- bins observations by $\hat{p}$
- compute observed proportion of 1s in each bin, compare to fitted $\hat{p}$ from model



Assessing binary predictions

Question: Logistic regression gives me $\hat{p}_i$. How do I turn $\hat{p}_i \in (0, 1)$ into $\hat{Y}_i \in \{0, 1\}$?

Thresholding :

$$\hat{Y}_i = \begin{cases} 1 & \hat{p}_i \geq c \\ 0 & \hat{p}_i < c \end{cases}$$

for some $c \in (0, 1)$

Confusion matrix

	$Y = 0$	$Y = 1$
$\hat{Y} = 0$	True Negatives (TN)	False Negatives (FN)
$\hat{Y} = 1$	False Positives (FP)	True Positives (TP)

Accuracy : $\hat{P}(\hat{Y} = Y) = \frac{TN + TP}{n}$

sensitivity : $\hat{P}(\hat{Y} = 1 | Y = 1) = \frac{TP}{TP + FN}$

specificity : $\hat{P}(\hat{Y} = 0 | Y = 0) = \frac{TN}{TN + FP}$

PPV : $\hat{P}(Y = 1 | \hat{Y} = 1) = \frac{TP}{TP + FP}$

NPV : $\hat{P}(Y = 0 | \hat{Y} = 0) = \frac{TN}{TN + FN}$

Confusion matrix

	No active disease	Active disease
FC < 61 mcg/g	15	7
FC ≥ 61 mcg/g	3	37

$$\text{Accuracy: } \frac{15+37}{62} = \frac{52}{62} \approx 0.839$$

$$\text{Sens: } \frac{37}{44} \approx 0.841$$

$$\text{Spec: } \frac{15}{18} \approx 0.833$$

$$\text{PPV: } \frac{37}{40} \approx 0.925$$

$$\text{NPV: } \frac{15}{22} \approx 0.682$$